# ALEJANDRO LUIS ARMAS BRAITHWAITE

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## EDUCATION

**University of Chicago** — Chicago, IL
*M.S. in Computational Analysis and Public Policy — GPA: 3.59/4.0* — **August 2026**
**Relevant Coursework:** *Machine Learning, Databases (SQL), Program Evaluation, Advanced Statistics, GIS*

**Pacífico Business School** — Lima, Peru
*M.A. in Economics* — **December 2019**

**Universidad del Pacífico** — Lima, Peru
*B.A. in Economics — Honors: Top Third of Students in 2015 and 2016* — **December 2017**

## RESEARCH PROJECTS

**Detecting Political Ideology in News Media Articles** — Chicago, IL
*Research Project, in collaboration with a faculty researcher* — **Summer 2025**

- Collected ~60,000 news articles and images (~10K per outlet) from 6 news media outlets using APIs, Selenium, and Requests.
- Cleaned and structured the multimodal (text + image) dataset for downstream NLP and computer-vision classification of political bias.

**Chicago Livability Score Project** — Chicago, IL
*University of Chicago* — **Winter 2025**

- Built crime, school, and environment indices for 60 Chicago ZIP codes from 257K crime records and 94K pollution complaints.
- Mapped point-level records to ZIP codes via spatial joins with Census shapefiles (GeoPandas, Shapely) and wrote unit tests (pytest) for a 4-person team's interactive Flask/Folium livability map.

**Predicting Dangerous Streets with ML** — Chicago, IL
*University of Chicago* — **Spring 2025**

- Built and compared Logistic Regression, Decision Tree, Random Forest, and Neural Network (MLP) classifiers to predict high-risk NYC street segments, using SMOTE to address class imbalance.
- Best model (tuned Random Forest) reached a 0.59 test F1 score vs. 0.36 for a random baseline and 0.49 for logistic regression, capturing non-linear effects of lane count, speed limits, and traffic calming.

## PROFESSIONAL EXPERIENCE

**The University of Chicago – Policy Lab (with IOM)** — Chicago, IL
*Policy Lab Consultant | Client: IOM* — **August 2025 – December 2025**

- Co-developed (5-person team) a standardized, Sendai-aligned Excel cost model estimating ~US$395M in annual investment needs for gender- and migration-inclusive Disaster Risk Management (DRM) across 5 South American countries.
- Adjusted GDP benchmarks with 4 normalized IDB governance and vulnerability indices, allocated costs across 5 policy dimensions, and validated findings with IOM and UN country offices.

**The University of Chicago – Harris School of Public Policy** — Chicago, IL
*Teaching Assistant – Machine Learning for Public Policy* — **August 2025 – December 2025**

- Led TA sessions, held office hours, and graded assignments and quizzes for a section of ~30 graduate students.
- Developed reproducible Python scripts and datasets implementing machine learning methods for public policy applications, supporting model replication and empirical analysis.

**Universidad del Pacífico – Faculty of Economics and Finance** — Lima, Peru
*Lecturer (2020–2025) | Teaching Assistant (2015–2019)* — **April 2015 – April 2025**

- Taught applied economics and econometrics courses for 9+ years, guiding students through empirical analysis with real-world datasets.

**Ministry of Foreign Trade and Tourism (Peru) – COPESCO** — Lima, Peru
*Public Investment Formulation Assistant* — **November 2023 – April 2024**

- Monitored evaluation and compliance for 10+ public tourism infrastructure projects, assessing proposals under Peru's Invierte.pe system.
- Cleaned and structured administrative datasets in Excel for feasibility analysis and descriptive statistical reporting.

**Ministry of Culture (Peru) – Economics Studies Unit** — Lima, Peru
*Data Consultant* — **October 2020 – February 2021**

- Consolidated 5 quantitative data sources and qualitative inputs into a centralized Excel dashboard for budget and policy analysis.
- Produced analytical tables and visualizations to support budget allocation decisions and policy briefs.

## TECHNICAL SKILLS

**Programming:** Python, SQL, R, Stata, JavaScript (D3.js)

**Python Libraries:** pandas, scikit-learn, GeoPandas, Shapely, Selenium, Altair, matplotlib

**Tools:** Git, Jupyter Notebooks, pytest, Excel (cost modeling, Pivot Tables, VBA, dashboards) | Languages: English (Fluent), Spanish (Native)

**Methods:** Machine learning, econometrics & causal inference, spatial analysis, web scraping & APIs, data visualization, reproducible research